

PolyAgora²

A Higher-Order Runtime Steering Architecture

CTO Spec Vo.1 – Runtime Governance Over Strategy Spaces

META-REGIME STEERING LAYER

QUANT ARCHITECTURE



CONTEXT

The Strategic Shift

PolyAgora² is not a refinement of a single alpha engine – it is a fundamental reframing of what the system is designed to do. The old model optimized for benchmark-relative performance. The new architecture optimizes for institutional robustness and future reachability across strategy manifolds.

Old Framing

- Beat the benchmark
- Replace Momentum as primary engine
- Single alpha strategy
- Point-in-time return maximization

New Framing

- Runtime governance architecture
- Strategy manifold steering
- Recoverability maximization
- Institutional robustness across regimes

Two-Level Architecture

The PolyAgora² system separates concerns into a three-tier stack. The meta layer governs; the strategy manifolds execute; the asset layer produces signals and returns. Dynamic allocation flows top-down – the governance layer shapes which strategy geometries are admissible at any given time.

1

Asset Layer

Signals · Weights · Returns · Execution

2

Strategy Manifolds

PolyAgora V7 · Momentum · Defensive / CHASH

3

Meta Governance Layer

M RTP Steering · Admissibility · Recoverability · Boundary Logic

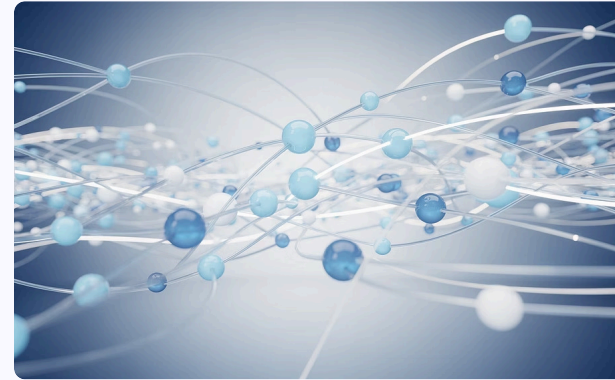
Strategy Manifolds: Complementary Geometries

The two primary strategy engines are not competitors – they are complementary operators covering different regions of the market geometry. The meta layer governs which is admissible under current regime conditions.



Momentum

- Strong in persistent, trending regimes
- Efficient exploitation of directional signals
- Vulnerable near regime reversals
- Exposed to fragmented market geometry



PolyAgora V7

- Strong during regime transitions
- Boundary-aware positioning
- Recoverability-oriented by design
- Handles fragmented market geometry



The objective is not to maximize immediate return.

The objective is to **maximize future reachable admissible states.**

CORE PRINCIPLE

FORMAL SPECIFICATION

Meta State Representation

The meta-regime state vector $\mathcal{R}_t$ encodes the four dimensions of regime geometry at time t . Each component is a continuous signal, enabling smooth steering decisions rather than binary regime switches.

$$\mathcal{R}_t = (r_t, v_t, c_t, b_t)$$

r_t

Regime persistence – how stable is the current market geometry?

v_t

Volatility stress – current realized and implied pressure levels

c_t

Admissibility confidence – posterior probability of current strategy validity

b_t

Boundary tension – proximity to admissibility boundary

FORMAL SPECIFICATION

M RTP Meta Score

The Meta-Regime Transition Probability score evaluates each candidate strategy manifold across three axes: current stability, future maneuverability, and expected damage upon misclassification. The weighting parameters α , β , γ are calibrated per regime class.

$$M RTP_i(t) = \alpha S_i(t) + \beta F_i(t) - \gamma D_i(t)$$

S_i – Stability

How well does strategy i preserve portfolio state under current geometry?
Weighted by α .

F_i – Future Maneuverability

Does strategy i preserve access to future admissible states? Weighted by β .

D_i – Expected Damage

What is the loss distribution if the current regime classification is wrong?
Subtracted, weighted by γ .

Meta Allocation Formula

Strategy weights are computed via a softmax transform over MRTP scores, gated by admissibility indicators **A_i(t)**. This produces smooth, probabilistic transitions – no binary switching, no cliff edges – governed by the temperature parameter λ .

$$W_i(t) = \frac{A_i(t) e^{\lambda MRTP_i(t)}}{\sum_j A_j(t) e^{\lambda MRTP_j(t)}}$$

Smooth Transitions

Softmax ensures gradual weight migration between manifolds as regime signals evolve.

Probabilistic Steering

Allocation reflects posterior confidence, not deterministic regime labels.

No Binary Switching

Admissibility gates provide hard floors without cliff-edge discontinuities.

Meta-Level Fixed Star Geometry

The meta-layer itself exhibits internal geometric structure. Rather than being a flat switch, it develops stable attractor regions – fixed stars – that represent reliable maneuverability centers under current geometry. Crucially, **meta fixed star $\neq$ best performing strategy**. It is the most stable maneuverability anchor.



Local Stars

Stable centers of maneuverability within the meta-manifold, serving as low-entropy anchors during high-uncertainty periods.



Manifold Blocks

Regions where specific strategy geometries are jointly admissible – overlap zones supporting regime-agnostic allocation.



Survival Topology

Boundary logic encoding which strategy combinations preserve institutional constraints through stress events.



⊗ Critical Constraint: Recursive Steering

The meta-layer must never become self-referentially recursive. It governs strategy manifolds – it must not govern itself.

Meta-Layer Discipline Requirements

The integrity of the governance architecture depends on maintaining strict separation between the meta-layer's operating cadence and the strategy manifolds beneath it. A meta-layer that adapts at the same frequency as its constituents collapses the two-level design.

Slower

Updates on daily or multi-day cadence, not intraday

Coarser

Operates on regime-level signals, not tick-level features

Lower Entropy

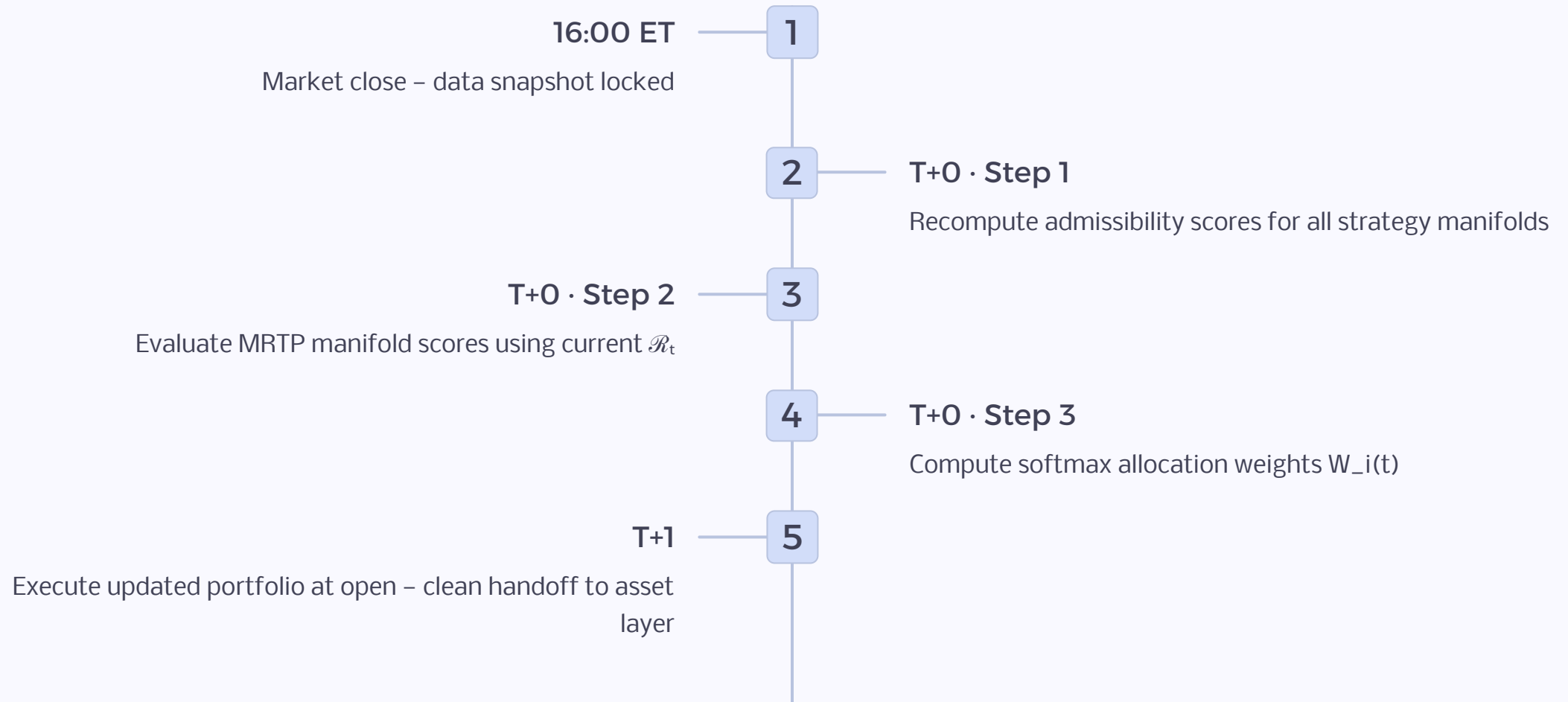
Fewer degrees of freedom – governance-grade simplicity

Non-Recursive

Cannot steer itself – separate validation scope required

Daily Runtime Flow

The end-of-day runtime cycle is deterministic and sequenced. All meta-layer computations are backward-looking, using only $\mathbf{F}_{(t-1)}$ signals to prevent hindsight contamination. Execution follows a fixed operational sequence after market close.





CHAPTER 2 · VALIDATION

Validation Protocol

Walk-forward only. No hindsight. Crisis-period stress testing required.

Validation Protocol Requirements

The validation framework enforces strict epistemic hygiene. Strategy manifolds are frozen during the meta-layer validation phase to isolate the governance signal from adaptive noise. Walk-forward testing is the only acceptable methodology – no in-sample optimization of meta-layer parameters.

1 Freeze Strategy Manifolds
Lock V7, Momentum, and Defensive configurations during meta-layer validation to isolate governance behavior from strategy adaptation.

2 Walk-Forward Validation Only
No lookahead. Meta-layer sees only **$F_{(t-1)}$** . All parameter selection must occur strictly out-of-sample.

3 Crisis Transition Stress Tests
Explicitly test performance through known regime fracture events: 2008, 2020, 2022. These are the moments governance architecture earns its value.

Validation Structure: Three-System Comparison

Validation is structured as a three-way comparison. Each system tests a distinct hypothesis. The meta-steered system must demonstrate superior recoverability and maneuverability – not merely superior average returns – to validate the governance thesis.

System	Objective	Validation Role
Static Momentum	Trend exploitation baseline	Upper bound during persistent regimes
Static PolyAgora V7	Governance baseline	Lower bound on admissibility cost
Meta-Steered System	Recoverability + maneuverability	Primary hypothesis under test

 Key metric: Does the meta-steered system demonstrate superior recovery speed and reduced drawdown depth during crisis transitions compared to either static baseline?

Meta Fixed Star does *not* mean "best performing strategy."

It means **stable maneuverability center under current geometry.**

KEY SCIENTIFIC DISTINCTION



Reference Polygon Architecture

The Reference Polygon (RP) provides the shared geometric language between layers. Maintaining a common RP across both the strategy and meta layers ensures semantic coherence – both layers are reasoning about the same underlying market geometry primitives.

Phase 1 · Initial Recommendation

- Use the same RP on both strategy and meta layers
- Maintain semantic coherence across layers
- Avoid ontology drift between governance and execution
- Simplest implementation path – reduce integration risk

Phase 2 · Long-Term Possibility

As the meta-layer matures, its RP may evolve toward more abstract governance primitives:

- **Persistence** – regime stability duration
- **Recoverability** – post-drawdown reachability
- **Convexity** – option value under uncertainty
- **Fragmentation** – coherence of signal geometry

Implementation Sequence

The build sequence is designed to minimize disruption to production systems. Each phase adds a governance layer without disturbing the strategy manifolds below. Validation is not a final step – it runs continuously from Phase 4 onward.

01	02	03
Keep V7 Intact No modifications to PolyAgora V7 during integration. It operates as a frozen strategy manifold.	Add Momentum Manifold Integrate Momentum as a parallel strategy engine with compatible signal and weight interfaces.	Add Defensive Manifold Integrate CHASH/Defensive as the low-entropy crisis-regime strategy option.
04	05	06
Build Admissibility Layer Implement the boundary logic and admissibility scoring against $\mathcal{R}_t$ for all manifolds.	Build MRTP Meta Steering Implement MRTP scoring and softmax allocation – the core governance compute layer.	Validate Walk-Forward Run three-system comparison using strict walk-forward protocol. No lookahead permitted.
07		
Stress-Test Crisis Periods Evaluate performance on 2008, 2020, 2022 regime fractures. Certify recoverability claims.		

Build Phases at a Glance

Governance Layer

Admissibility rules and MRTTP policies added.

Validation

Walk-forward and crisis stress tests only.

Foundation

V7 intact; add manifolds and baseline infra.

Production Release

Certified meta-steered system goes live.

Each phase has explicit entry and exit criteria. The governance layer is not deployed until walk-forward validation is certified clean – no production shortcuts during stress-test phases.

System Design Principles

1

Separation of Concerns

Governance layer and execution layers operate on distinct cadences, feature sets, and optimization objectives.

2

Smooth Probabilistic Allocation

No binary regime switches. Softmax-gated MRTP scores produce continuous, auditable weight transitions.

3

Recoverability as Primary Objective

The system is optimized for future reachable admissible states – not point-in-time return maximization.

4

Epistemic Hygiene in Validation

Walk-forward only, frozen manifolds, crisis-period stress tests – zero hindsight contamination tolerance.

A New Category

PolyAgora is no longer merely a strategy engine.

It becomes **a runtime governance architecture over strategy spaces** – steering admissibility, preserving recoverability, and maintaining institutional robustness across all market regimes.

POLYAGORA² · CTO SPEC Vo.1

META-REGIME STEERING LAYER

